Stoplight and Pedestrian Signal Lab

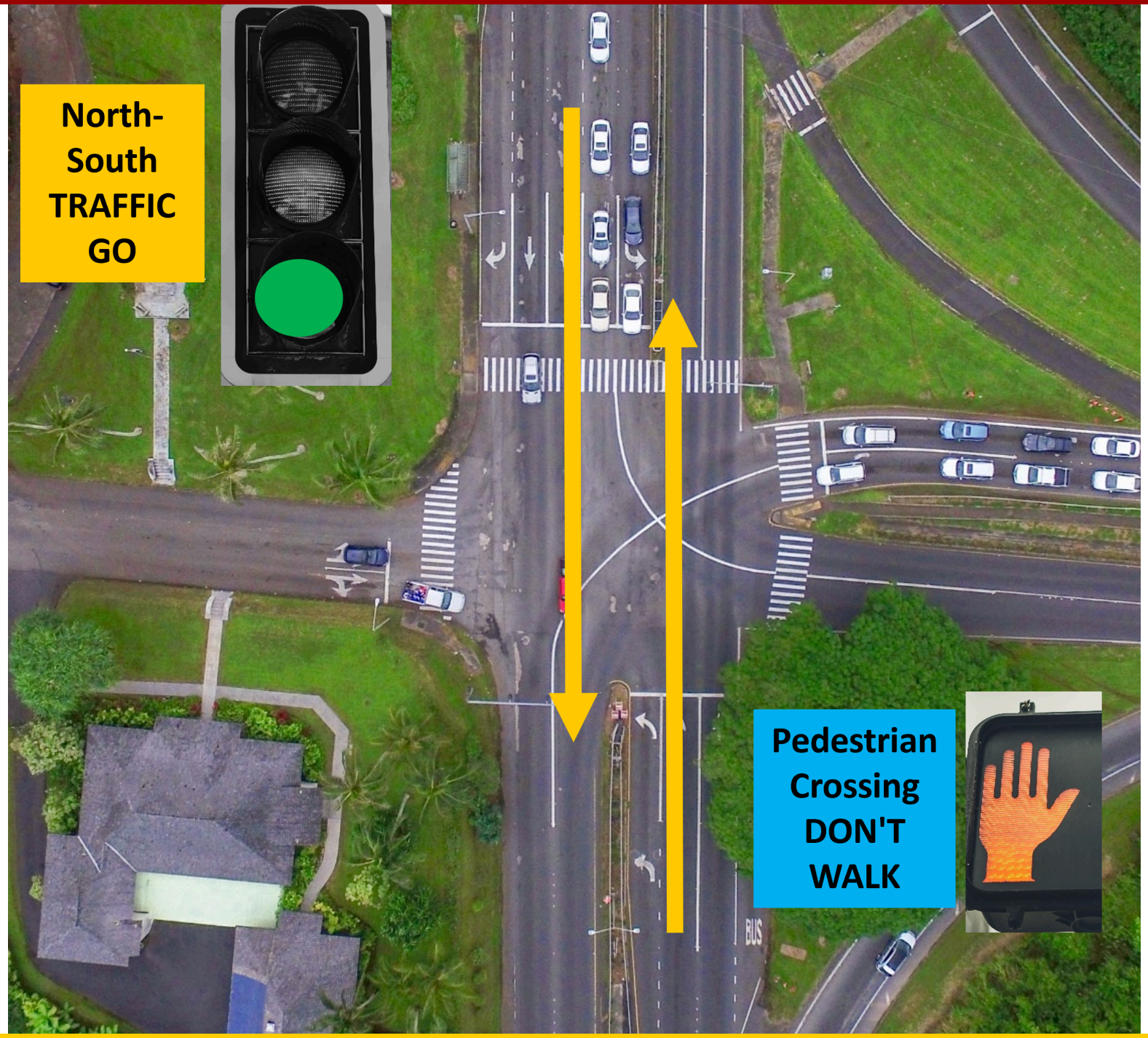
TAC 348 - Making Smart Devices

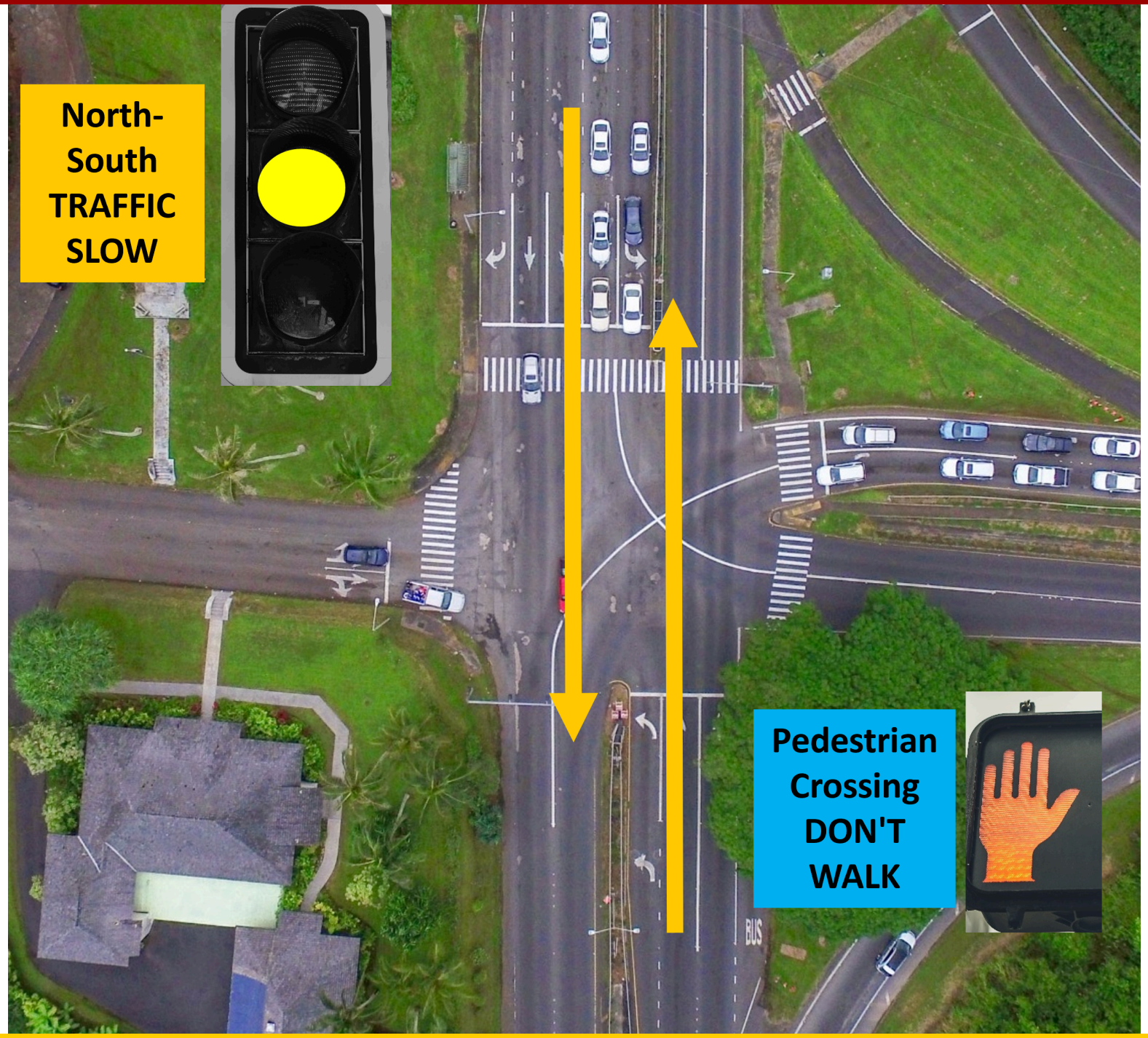
Lab Learning Objectives

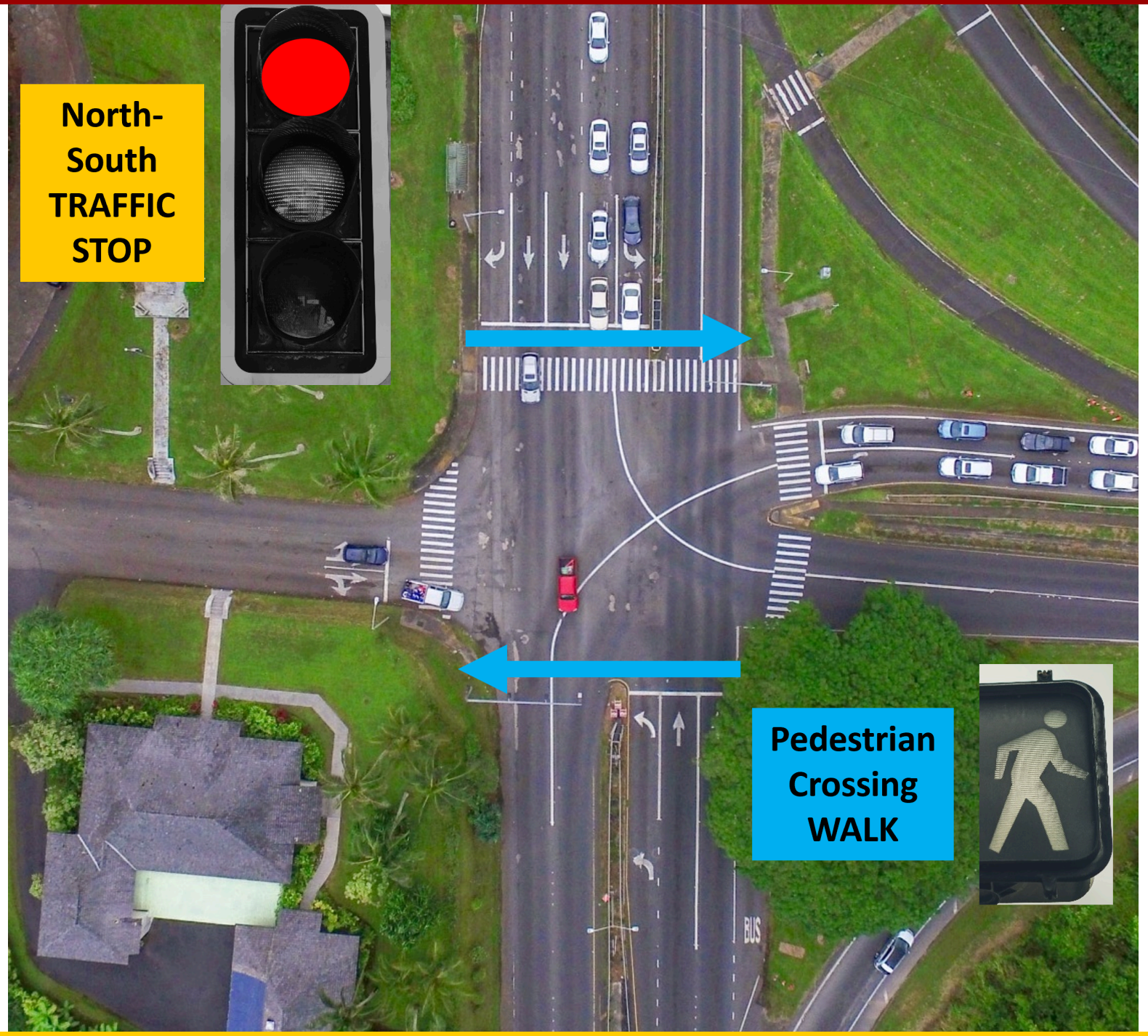
- Model a traffic intersection as a state machine using enum values and state variables
- Use millis() to create non-blocking timed state transitions
- Control two traffic signals and pedestrian LEDs
- Control an independently blinking "Don’t Walk"

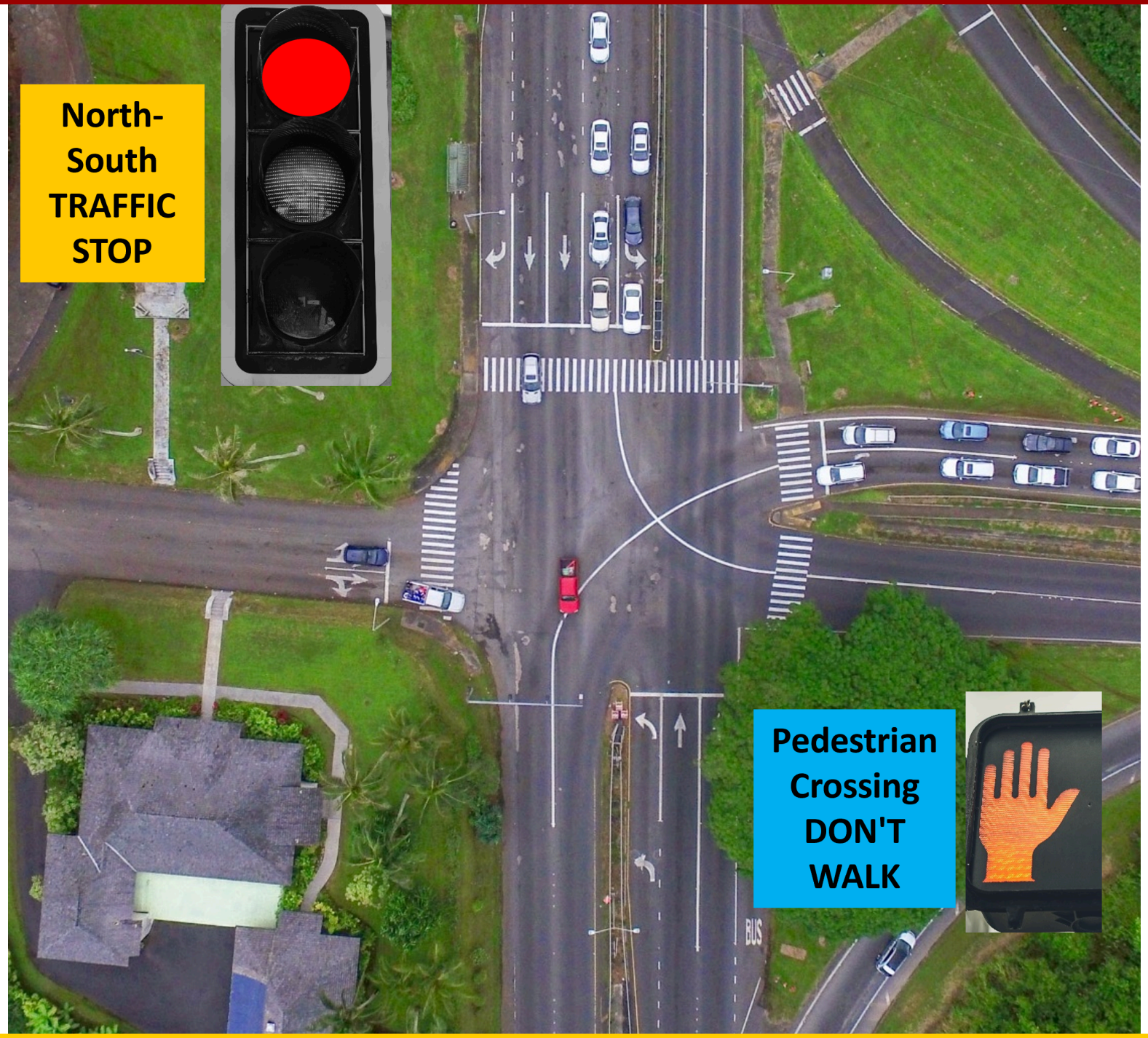
In class Lab - Stoplight and Pedestrian Signal

- We are going to build a stoplight and pedestrian signal for an intersection.
- Car traffic goes North-South.
- Pedstrian traffic goes East-West
- Green lights, red lights, and the walk signal will stay on for 5 sec (`LONG_LIGHT_DURATION`)
- Yellow lights and the don't walk signal will stay on for 2 sec (`SHORT_LIGHT_DURATION`)
- Here is how this type of intersection works


North-
South
TRAFFIC
GO
Pedestrian
Crossing
DON'T
WALK
BUS


North-
South
TRAFFIC
SLOW
Pedestrian
Crossing
DON'T
WALK
BUS


North-
South
TRAFFIC
STOP
Pedestrian
Crossing
WALK
BUS


North-
South
TRAFFIC
STOP
Pedestrian
Crossing
DON'T
WALK
BUS

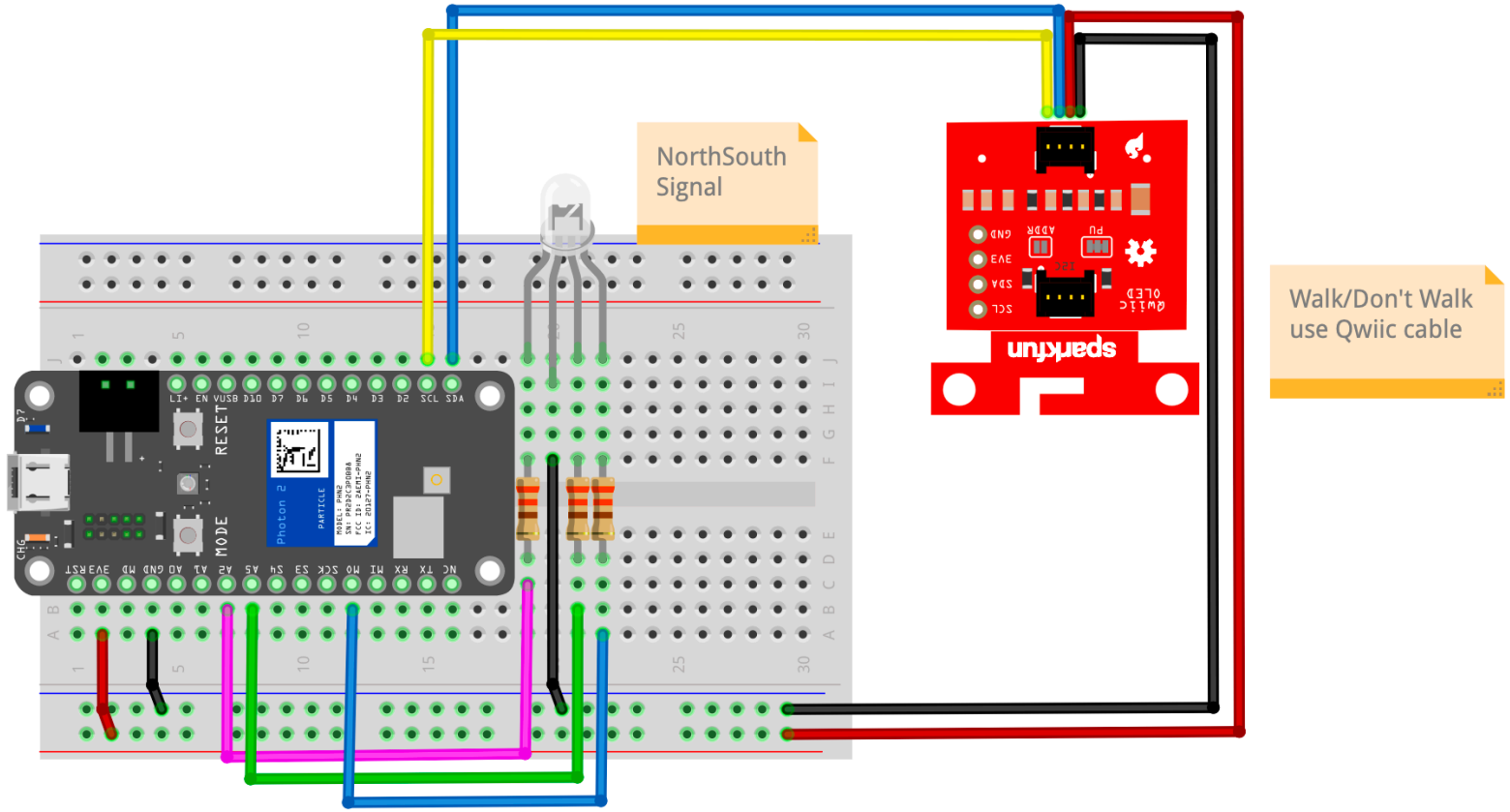
In class Lab - Stoplight and Pedestrian Signal

- Download starting code
  - Go to https://tinyurl.com/ProjectZipGH
  - Paste the following link into the top right

```
https://github.com/reparke/TAC348-Making-Smart-Devices/tree/main/_exercises/week08/stop_light_rgb_leds_oled_start
```

- Wire device as shown below

Wiring Diagram



Components

- 1 RGB LED
- MicroOLED
- 3 x 330 Ohm resistors

Wiring Pin Guide

RGB LED	Photon 2 Pins
NorthSouth RGB LED - Red	A2
NorthSouth RGB LED - Green	A5
NorthSouth RGB LED - Blue	D15 (MO)

MicroOLED (Qwiic cable)	Photon 2 Pins
GND	GND
3V3	3V3
SDA (Blue)	SDA
SCL (Yellow)	SCL

Lab Stages

1. Create cycle with only traffic (North-South) stop lights
2. Add Pedestrian Walk / Don't Walk (East-West) message to OLED
3. Make Don't Walk light blink when it is on (if time)

- We will work through stage 1 together

General Approach

- We will need to create a way to represent the states using `enum`
- We will need to track state transitions with **variables** for state
- We will need to control timing with `millis()`
- What does `loop()` do
  - calculate new state
  - output traffic signal LEDs based on state
  - independently flash walk signal *(later)*

Credits

- Bill Siever - CS132 FSM (Washington University at St. Louis)
- Bill Siever - CS132 Delta Timing (Washington University at St. Louis)
- USC EE109 State Machines
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